

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

The release of this audit program to parties external to IBM requires the written approval of the Chief Auditor.

This program is intended as a guide for assessing risk management in this process through the performance of an audit or a similar engagement. Risk management is the process to identify, manage, and control potential events in order to provide reasonable assurance related to the achievement of IBM's business and control objectives. IBM's control objectives fall into five broad categories:

1. compliance with external laws and regulations, as well as IBM policies and procedures
2. accomplishment of business objectives
3. reliable information for financial reporting and decision making
4. efficient operations, and
5. safeguarding both intellectual and physical assets.

The control assessment questions are intended to address process objectives from the categories above and the risks to achieving those objectives. Objectives and risks in the actual process under review may differ based on organizational, operational, regulatory, and economic factors. As such, individuals performing the engagement must formulate a test strategy based on risks relative to the actual process under review, regardless of whether the risks are identified within this program.

DATE CHANGED	SECTION #	SUBTITLE IN SECTION	CHANGED BY (NAME)	COMMENTS
December 2024	Planning, Control Assessment Question	RAE, Reference material, Counterparty Risk	Adriana Lippi	Updated RAE risk value, FINs listing, responsibilities and timeline for counterparty risk and limits
May 2023	All	All	Jacob Nyamwega	Replaced GBDB with new management tool FIS-eBAM
May 2023	Control Assessment Question	9. Counterparty Risk Evaluation	Jacob Nyamwega	Changed CPL) process to monthly reporting from bi-weekly reporting
February 2023	All	All	CA&AS	Converted from Notes into a Word document and into a Box folder

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

Table of Contents	
PROCESS AND OBJECTIVE	4
PLANNING: Audit Program Risk Alignment.....	5
PLANNING: Data Analysis.....	6
PLANNING: Advanced Sampling.....	8
PLANNING: Reference Material	9
PLANNING: Glossary of Terms and Definitions	10
PLANNING: Fraud Template	14
CONTROL ASSESSMENT QUESTIONS	16
Key Control Question: 1. Cash Management.....	16
Primary Test: Bank Relations (CM001.P)	17
Primary Test: Cash Flow Forecasting (CM002.P)	19
Secondary Test: Wire Transfer (CM003.S)	20
Primary Test: Bank Reconciliations (CM004.P)	21
Secondary Test: Cash Funds and Blank Checks (CM005.S)	22
Secondary Test: Cash Receipts (CM006.S)	23
Secondary Test: Cash Transactions (CM007.S)	24
Key Control Question: 2. Investment Portfolio.....	25
Primary Test: Valuation and Control of Portfolio (IP001.P)	27
Primary Test: Cash and Investment Planning (IP002.P)	29
Primary Test: Broker/Dealer Relationships (IP003.P)	30
Primary Test: Interest Income (IP004.P)	31
Key Control Question: 3. Borrowing Management.....	32
Primary Test: Valuation and Control of Debts (BM001.P)	33
Primary Test: Cash and Debt Planning (BM002.P)	34
Secondary Test: Payment of Debts (BM003.S)	35
Primary Test: Interest Expenses on Debts (BM004.P)	36
Key Control Question: 4. Intercompany Transactions	37
Primary Test: Settlement of Intercompany Charges (IT001.P)	38
Key Control Question: 5. Foreign Exchange Management	39
Primary Test: Foreign Exchange Transactions (FX001.P)	40
Primary Test: Currency Exposures (FX002.P)	41
Primary Test: Accounting Controls (FX003.P)	42
Key Control Question: 6. Investment in Derivatives	43
Primary Test: Investment in Derivatives (ID001.P)	44

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

Primary Test: Valuation of Derivatives (ID002.P).....	46
Primary Test: Financial Accounting Standard 133 (ID003.P)	47
Key Control Question: 7. Policies and Procedures.....	48
Corporate and Regional Policies and Procedures (PP001.P)	49
Key Control Question: 8. Management Reporting.....	51
Primary Test: Management Reporting and Review (MR001.P)	52
Key Control Question: 9. Counterparty Risk Evaluation	53
Primary Test: Counterparty Risk (CR001.P).....	54
Key Control Question: 10. Disbursements	56
Primary Test: Procurement Bypass (DS001.P)	57
Key Control Question: 11. Acquisition Integration	58
Primary Test: Acquisition Integration (AI001.P).....	59
Key Control Question: 12. Share Repurchases.....	60
Primary Test: Share Repurchases (SR001.P)	61
Key Control Question: 13. Bank Guarantee	62
Primary Test: Bank Guarantee Management (BG001.P)	63
Key Control Question: 14. Separation of Duties	64
Primary Test: Separation of Duties (SD001.P)	65
Key Control Question: 15. Business Continuity Planning.....	66
Secondary Test: Business Continuity Planning (BC001.S)	67

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

PROCESS AND OBJECTIVE

Treasury operations within IBM and its subsidiary companies encompasses process which relate to cash receipts and disbursements, investments and borrowing, and the safeguarding of company funds. The Risk Management process in the treasury function starts when a decision is made in the risk management function to eliminate or reduce an existing financial risk for IBM. The process includes ongoing assessment, monitoring of the risk minimization transactions, and the underlying risks.

The processes include the following areas or processes:

1. Cash Management
2. Investment Portfolio
3. Borrowing Management
4. Intercompany Transactions
5. Foreign Exchange Management
6. Investment in Derivatives
7. Policies and Procedures
8. Management Reporting
9. Counterparty Risk Evaluation
10. Disbursements
11. Acquisition Integration
12. Share Repurchases
13. Bank Guarantees
14. Separation of Duties (SOD)
15. Business Continuity Planning

In certain countries, treasury function may also have the responsibility for other activities such as Risk and Insurance Management, Taxes, Retirement Fund, and Alliances. Each of these activities is covered by a separate audit program.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

PLANNING: Audit Program Risk Alignment

Treasury Operations

Defined Risk Area	Audit Risk Level	Key Control Question	Test Number
Cash Management and Forecast	High	Cash Management (CM)	CM002.P, CM004.P, CM005.S, CM006.S
Investment and Borrowing	High	Investment Portfolio (IP) Borrowing Management (BM) Policies and Procedures (PP) Management Reporting (MR)	IP001.P, IP002.P, IP004.P BM001.P, BM002.S, BM003.S, BM004.P PP001.P MR001.P
Foreign Exchange (FX) and Hedging	High	Foreign Exchange Management (FX) Investment in Derivatives (ID) Policies and Procedures (PP) Management Reporting (MR)	FX001.P, FX002.P, FX003.P ID001.P, ID002.P, ID003.P PP001.P MR001.P
Bank Guarantees	High	Bank Guarantees (BG)	BG001.P
Disbursements, Manual Payments, and Factoring	Medium	Cash Management (CM) Intercompany Transactions (IT) Share Repurchases (SR)	CM003.S, CM005.S, CM007.S IT001.P SR001.P
Bank Administration, Data Protection, and Bank Fees	Medium	Cash Management (CM) Investment Portfolio (IP)	CM001.P IP003.P
Acquisitions	Medium	Acquisition Integration (AI)	AI001.P
Accounting Reconciliations	Medium	Cash Management (CM)	CM004.P
Counterparty Limits	Medium	Counterparty Risk Evaluation (CR)	CR001.P
Cybersecurity and Payment Systems	Medium	Cash Management (CM)	CM001.P, CM003.S
Business Continuity	Low	Business Continuity Planning (BC)	BC001.S

TREASURY OPERATIONS AUDIT PROGRAM #12-00-00

PLANNING: Data Analysis

Data analysis should be a part of every engagement where data is available in databases or data warehouses that can be accessed through query programs. This concept enhances the risk-based auditing approach, makes the audit process more efficient, and increases the productivity of the audit team. When the audit team does not have the skills required to perform these tasks, assistance should be sought from the Technology team.

All data used for analysis must be reconciled to an independent source so that the auditor has confidence that the universe is complete.

The [Data Analysis engagement procedures in the Audit Manual](#) (link internal to CA&AS members only) provide further suggestions.

This table shows what data can be reviewed and analysis techniques.

Data to be Analyzed	Analysis to be Performed	Possible Anomalies	Concern
Bank Accounts	Ask IT team to pull bank accounts in ledger and compare to bank accounts in FIS-eBAM.	- Accounts closed in FIS-eBAM have not been closed in ledger and vice versa. - Accounts opened in FIS-eBAM have not been opened in ledger and vice versa.	- For accounts in FIS-eBAM and not in ledger, cash transactions may not be recorded appropriately. - For accounts in ledger and not in FIS-eBAM, could be a FIN181 violation.
Checks	- Total number and value of checks issued/received in logs. - Safe room logs.	- Checks to inconsistent vendors. - Sequential checks not logged. - Logs are not filled out appropriately.	- Vendors that are only used once/twice could have the potential to be "fraudulent". - Missing checks or sequence of numbers could indicate fraudulent transactions. - Safe room entering/leaving not properly documented could represent that controls around the checks are not adequate.
Bank fees	Analyze type and dollar amount of bank fees month-to-month with the same banks.	Bank fees vary month-to-month for the same type of transactions/services.	Charged bank fees could be incorrect.

TREASURY OPERATIONS AUDIT PROGRAM #12-00-00

PLANNING: Data Analysis (Continued)

Data to be Analyzed	Analysis to be Performed	Possible Anomalies	Concern
Debt and Investments	Analyze universe of debt and investments, look at frequency, amounts, maturity dates, and rates.	- Multiple investments on the same date or dates close together. - No business justification for debt (e.g., not supported by cash forecast).	- Excess transactions fees for investments (unless CPL issue explains usage for multiple banks). - No business need for debt, could result in unnecessary financial exposure and costs.
Hedging Transactions	- Analyze the type (swap, forward, etc.), frequency, and amount of transaction. - Pivot each day to have the total amount hedged.	- Type of transaction is not allowed according to IBM policies/instructions. - Total hedges exceed the allowable delegation.	- Instrument utilized is outside of the allowed transactions per IBM policies/instructions. - Inappropriate hedging activity outside of allowable delegation.

TREASURY OPERATIONS AUDIT PROGRAM #12-00-00

PLANNING: Advanced Sampling

Per earlier communications with the Treasury management team, sample requests require significant time commitment from the client team. Therefore, Treasury is willing to have additional universe requests in the Initial Data Request (IDR) and audit has agreed to send out sample requests in advance. The timing of the sample requests to due dates is outlined in the table below, with some before opening conference and others during process interview week. For each engagement, depending on the universes of data, there may need to be adjustments to which sample requests are sent to allow each auditor to receive responses to one scope area on the first Friday. In all cases, additional samples may be required after process interviews to capture unique requirements of the auditable unit. The audit team leader and client focal will work out the detailed timeline for those areas deemed relevant to the unit being audited.

Taxonomy		Audit Program in workpapers		IDR in Community	Outline of Early Testing Timing		
Defined Risk Area	Risk Level	Key Control Question	Test Number	IDR numbers and descriptions	Issue the RFI on:	# of days to respond	Due
Cash Management and Forecast	High	Cash Management (CM)	CM002.P, CM004.P, CM005.S, CM006.S	B04 - Cash and Debt Positions (due 1st) B06 - Unexpected Overdrafts (due 2nd) B09 - Intercompany (due 2nd) B14 - Cash Forecasting (due 2nd) B15 - Intercompany Settlements (due 3rd) B16 - Dividends or Equity Reductions (due 3rd)	Thursday before opening conference	6	Friday of week 1
Investment and Borrowing	High	Investment Portfolio (IP) Borrowing Management (BM) Policies and Procedures (PP) Management Reporting (MR)	IP001.P, IP002.P, IP004.P BM001.P, BM002.S, BM003.S, BM004.P PP001.P	B05 - Borrowing and Debt (due 2nd) B07 - Investments and Derivatives (due 2nd)	Friday before opening conference	6	Monday of week 2
Foreign Exchange and Hedging	High	Foreign Exchange Management (FX) Investment in Derivatives (ID) Policies and Procedures (PP) Management Reporting (MR)	FX001.P, FX002.P, FX003.P ID001.P, ID002.P, ID003.P PP001.P MR001.P	B08 - Foreign Exchange Transactions (due 2nd) B10 - Hedging - N/A for most (due 2nd)	Thursday of week 1	3	Tuesday of week 2
Bank Guarantees	High	Bank Guarantees (BG)	BG001.P	B11 - Bank Guarantees (due 2nd)	Thursday before opening conference	6	Friday of week 1
Disbursements, Manual Payments, and Factoring	Medium	Cash Management (CM) Intercompany Transactions (IT) Share Repurchases (SR)	CM003.S, CM005.S, CM007.S IT001.P SR001.P	B13 - Disbursements, Manual Payments, and Factoring (due 3rd)	Likely included in Cash Management sampled dates, if not, select Thursday week 1	3	If applicable, due Tuesday of week 2
Bank Administration, Data Protection, and Bank Fees	Medium	Cash Management (CM) Investment Portfolio (IP)	CM001.P IP003.P	B19 - Banking Relationships (due 2nd) B18 - Bank Fees (due 3rd) B20 - Positive Bank Confirmations (due 3rd)	Thursday before opening conference. Note positive confirmations still in IDR.	6	Friday of week 1
Acquisitions	Medium	Acquisition Integration (AI)	AI001.P	B17 - Acquisitions (due 3rd)	Friday before opening conference	6	Monday of week 2
Accounting Reconciliations	Medium	Cash Management (CM)	CM004.P	C03 - Account Reconciliations (due 2nd)	Auditors have access and send FYI RFI in Week 2	-	N/A
Counterparty Limits	Medium	Counterparty Risk Evaluation (CR)	CR001.P	B12 - Counterparty Limit (due 2nd)	Thursday of week 1	3	Tuesday of week 2
Cybersecurity and Payment Systems	Low	Cash Management (CM)	CM001.P, CM003.S	A04 - Personal Information and Data Protection (due 1st) D01 - Application Listing (due 1st)	Provided in IDR, and generally no samples.	-	Samples would be requested onsite if tested, with a 24-48 hour due
Business Continuity	Low	Business Continuity Planning (BC)	BC001.S	E03 - Disaster Recovery Plan (due 4th)	Provided in IDR, no samples required.	-	N/A

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

PLANNING: Reference Material

In the IDR, you will receive the latest version of the IBM Global Treasury Policies with exceptions that could pertain to the unit being audited. This will be a primary reference for the engagement.

[Finance Corporate Instructions](#) applicable for Treasury Operations, **with those in bold signed by a Treasury officer.**

- FIN101: Control of disbursements
- **FIN160: Capital structure of IBM subsidiaries and other organizational structures**
- **FIN177: Parent Guarantee Fee Management**
- FIN179: Quarterly Certification of Management
- **FIN180: IBM's Worldwide Payment Terms**
- **FIN181: Bank account management**
- FIN182: Management Self Assessment of Controls, inclusive of Appendix C on Separation of Duties
- **FIN185: Formation and maintenance of legal entities (signed jointly by Treasury and Legal)**
- FIN187: Clip Level Establishment and Revalidation
- **FIN191: Entering into Leasing Transactions (IBM as a Lessee)**
- **FIN 192: Use of Restricted Currencies**
- **FIN 194 Intercompany Transactions Policy and Procedures**
- GDPR: General Data Protection Regulation Readiness Community
- Business Continuity: BPBC127: Business Process Business Continuity planning

[GLOBAL BCM - BCM inside IBM: https://w3.ibm.com/w3publisher/global-bcm/bcm-inside-ibm](https://w3.ibm.com/w3publisher/global-bcm/bcm-inside-ibm)

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

PLANNING: Glossary of Terms and Definitions

Arbitrage: simultaneous purchase and sale of the same asset to take advantage of price differences prevailing on different markets. A pure arbitrage involves effectively trading of the same or similar instruments for different prices at the same time.

Basis Points (BPS): common unit of measure for interest rates and other percentages in finance. One basis point is equal to 1/100th of 1%, or 0.01% (0.0001), and is used to denote the percentage change in a financial instrument. The relationship between percentage and basis points can be summarized as: 1% = 100 basis points, and 0.01% = 1 basis point.

Counterparty: a principal party to a financial transaction, other than an intermediary. When looking from the buyer's viewpoint, the seller is the counterparty and vice versa.

Credit (debt) rating: An assessment of the creditworthiness of a borrower in general terms or with respect to a particular debt or financial obligation. A credit rating can be assigned to an individual, corporation, state or provincial authority, or sovereign government. Credit assessment and evaluation for companies and governments is generally done by a credit rating agency such as Standard & Poor's, Moody's, or Fitch. Debt obligations would typically not be assigned a rating better than the country rating.

Dealer: a person or firm acting as a principal party in buying and selling securities for their own account.

Internal Rate of Return (IRR): a metric used to measure the profitability of an investment. The IRR, when used as a discount rate, equates the price of a financial instrument with its individual discounted cash flows.

Leverage: the investment strategy of using borrowed money. Specifically, the use of various financial instruments or borrowed capital to increase the potential return of an investment. Leverage can also refer to the amount of debt used to finance assets. When one refers to something (a company, a property or an investment) as "highly leveraged", it means that item has more debt than equity.

Market price: the current price which an asset has been bought or sold (usually the last reported price).

Plain vanilla: the most basic or standard version of a financial instrument, usually options, bonds, futures and swaps. Plain vanilla is the opposite of an exotic instrument, which alters the components of a traditional financial instrument, resulting in a more complex security.

Sensitivity analysis: technique used to determine how different values of an independent variable impact a particular dependent variable under a given set of assumptions, e.g. if interest rates and prices move up and down by 100, 200, and 300 basis points. This technique is used within specific boundaries that depend on one or more input variables, such as the effect that changes in interest rates have on bond prices.

Settlement date: the date on which security transactions are settled by delivering or receiving securities and receiving or paying cash pursuant to an earlier agreement of purchase or sale called trade.

SONIA (Sterling Overnight Index Average): an interest rate benchmark – also known as a reference rate or a benchmark rate. It is based on the terms of actual overnight transactions and so the rate is largely unaffected by bank credit risk.

Trade: an agreement of purchase and sale in a securities market, to be settled or performed by payment and delivery on a later settlement date.

Trade date: the date when an order to purchase, sell, or otherwise acquire a security is executed, to be settled or with a later settlement date.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

PLANNING: Glossary of Terms and Definitions (Continued)

Trading book: portfolio of financial instruments held by a financial institution. Financial instruments in a trading book are purchased or sold for reasons including to facilitate trading for the institution's customers, to profit from trading spreads between the bid and ask prices, or to hedge against various types of risk.

Yield: the income returns on an investment, such as the interest or dividends received from holding a security. The yield is usually expressed as an annual percentage rate based on the investment's cost, current market value or face value.

Yield curves: a line that plots the interest rates, at a set point in time, of bonds having equal credit quality but differing maturity dates.

Yield to Maturity (YTM): is the internal rate of return of an investment if the investor holds the asset until maturity and if all payments are made as scheduled.

Derivatives, Forwards, and Swaps

Currency swap (or cross-currency swap): involves the exchange of interest and sometimes of principal in one currency for the same in another currency. Interest payments are exchanged at fixed dates through the life of the contract.

Derivative: a security with a price that is dependent upon or derived from one or more underlying assets. Its value is determined by fluctuations in the underlying asset. The most common underlying assets include stocks, bonds, commodities, currencies, interest rates and market indexes.

Forward: a contract between two parties to buy or sell an asset at a specified price on a future date.

Forward exchange contract or forward contracts: agreements between two parties to exchange two designated currencies at a specific time in the future.

Forward Rate Agreement (FRA): is a contract between parties that determines the rate of interest, or the currency exchange rate, to be paid or received on an obligation at a future date. FRAs are cash settled with the payment based on the net difference between the interest rate and the reference rate in the contract.

Future: financial contract obligating the buyer to purchase an asset or the seller to sell an asset, such as a physical commodity or a financial instrument, at a predetermined future date and price.

Hedge: an investment to reduce the risk of adverse price movements in an asset. Normally, a hedge consists of taking an offsetting position in a related security, such as a futures contract.

International Swaps and Derivatives Association (ISDA): standard document that is commonly used to govern over-the-counter derivatives transactions. Outlines the standard terms to be applied to a derivatives transaction between two parties.

Mark to Market (MTM): measure of the fair value of accounts that can change over time, such as assets and liabilities. The accounting act of recording the price or value of a security, portfolio or account to reflect its current market value rather than its book value. MTM aims to provide a realistic appraisal of an institution's or company's current financial situation.

Notional or notional principal amount: in an interest rate swap, is the predetermined dollar amounts on which the exchanged interest payments are based.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

PLANNING: Glossary of Terms and Definitions (Continued)

Over-the-Counter (OTC): a security traded directly between two parties in some context other than on a formal exchange such as the New York Stock Exchange (NYSE), Toronto Stock Exchange, or the NYSE MKT, and usually through a dealer network.

Plain vanilla swap: one of the simplest financial instruments contracted in the over-the-counter market between two private parties, both of which are usually firms or financial institutions. Most commonly used to describe an interest rate swap in which a floating interest rate is exchanged for a fixed rate or vice versa.

Swap: a derivative contract through which two parties exchange financial instruments. These instruments can be almost anything, but most swaps involve cash flows based on a notional principal amount that both parties agree to. Usually, the principal does not change hands. One cash flow is generally fixed, while the other is variable, that is, based on a benchmark interest rate, floating currency exchange rate or index price.

Swaption (swap option): the option giving the holder the right, but not the obligation, to enter into an interest rate swap or some other type of swap on a specified future date, in exchange for an option premium.

Options

American option: an option that can be exercised any time during the exercise period.

Call: an option contract that gives the owner the right, but not the obligation, to buy a specified amount of an underlying security at a specified price within a specified time.

European option: an option that can only be exercised at the end of its life, at maturity date.

Option: a contract between two parties, which gives one party the right, but not the obligation, to buy or sell an asset, currency, or rate for a specified price.

Put: an option contract that gives the owner the right, but not the obligation, to sell a specified amount of an underlying asset at a set price within a specified time.

Risks

Basis risk: imperfect correlation between two offsetting investments that creates the potential for excess gains or losses in a hedging strategy, thus adding risk to the position.

Counterparty risk: the risk to each party of a contract that the counterparty will not honor its contractual obligations. It includes both settlement and credit risks.

Credit risk: the risk of loss arising from the default of a counterparty, or from the counterparty failing to meet all contractual obligations as defined and on time. This risk is most prevalent in commercial and consumer lending.

Currency risk: the risk that changes in the value of foreign currencies will affect the value of an asset, liability, or financial instrument that is denominated in the foreign currency. Investors or companies that have assets or business operations across national borders are exposed to currency risk that may create unpredictable profits and losses. Currency risk can be reduced by hedging, which offsets currency fluctuations.

Fraud risk: the risk related to intentional wrongdoing. Risk of an intentionally deceptive action designed to provide the perpetrator with an unlawful gain, or to deny a right to a victim.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

PLANNING: Glossary of Terms and Definitions (Continued)

Interest rate risk: the risk that an investment's value will change due to a change in the absolute level of interest rates, in the spread between two rates, in the shape of the yield curve, or in any other interest rate relationship. Such changes usually affect securities inversely and can be reduced by diversifying (investing in fixed-income securities with different durations) or hedging (such as through an interest rate swap).

Legal risk: the financial risk that stems from a specific contractual obligation, or the lack thereof, pertaining to a transaction entered by IBM.

Liquidity risk: the risk that available cash will be inadequate to meet short-term obligations. The risk stemming from the lack of marketability of an investment that cannot be bought or sold quickly enough to prevent or minimize a loss.

Market risk or systematic risk: the possibility for an investor to experience changes in their book or market value of a financial instrument, commodity, or other asset or liability, due to factors that affect the overall performance of the financial markets in which he is involved.

Operational risk: the risks a company undertakes when it attempts to operate within a given market or activity. Results from breakdowns in internal procedures, people, or systems.

Regulatory risk: the risk that a change in laws and regulations will materially impact a security, business, sector or market.

Risk assessment: is a general term used to determine the likelihood of loss on a particular activity or transaction. The process of assessing risk helps to determine if an investment is worthwhile, what steps may be taken to mitigate the risks.

Risk identification: the analysis of the risks that different financial instruments pose, individually and as a group, to the company.

Settlement risk: the risk that one party will fail (default) to deliver the terms of a contract with another party at the time of settlement.

Sovereign risk: the risk that a foreign nation will either fail to meet debt repayments or not honor sovereign debt payments.

Speculative risk: any risk that is engaged in for its own sake (as conscious choice) and is not a business risk, including transactions engaged in solely for the purpose of profiting from market movements. The risk is unrelated to production and delivery of products and services and could be avoided without a substantial economic penalty to IBM. Speculative risk also includes normal market making where the expectation is simply to profit from bid-ask differences.

Systematic risk: see market risk.

Transaction risk: risk associated with the time delay between entering into a contract and settling it.

Translation risk: risk associated with companies that deal in foreign currencies or list foreign assets on their balance sheets. Typically, the excess of foreign-currency assets over foreign-currency liabilities of foreign subsidiaries. Translation gains result from increases in the value of the foreign currency and losses in the event of decreases.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

PLANNING: Fraud Template

Fraud is any intentional act or omission designed to deceive others and resulting in the victim suffering a loss and/or the perpetrator achieving a gain. Fraud can be categorized into fraudulent financial reporting, misappropriation of assets, and improper or unauthorized expenditures.

Description of Fraud Risk	Typical Controls that should Detect the Fraud Risk	Testing Approach
Blank Checks	• Check inventory log • Limited access to safe • Management review of checks inventory • Bank seal check	• Perform a blank checks inventory and count. • Determine who has access to cash, blank checks, bank seals, and cash receipts for separation of duties concerns and valid business need. • Review the process to discard blank checks. • Review the blank check log and ensure management has review it for accuracy. • Request the list of authorized employees who can access the safe and reconcile against the log and check inventory for accuracy. • Validate what efforts have been made to minimize the use of checks.
Unauthorized or Unsupported Disbursements	• Disbursement Approvals • Separation of Duties • Access Management Controls	• Review supporting documentation to validate proper business need and approvals. • Review the disbursement process to ensure compliance with the required separation of duties. • Review the access management procedures for valid business needs.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

PLANNING: Fraud Template (Continued)

Description of Fraud Risk	Typical Controls that should Detect the Fraud Risk	Testing Approach
Bank Accounts	• Bank account revalidation • Positive bank confirmation • Review Signatories	• Ensure that there was a valid business justification and supporting documentation to open an account. • Select a sample of banks and ensure that they were notified timely about changes. • Ensure that the annual bank certification was performed. • Request a positive bank confirmation with some banks IBM has relationships with. • Reconcile that list with the bank accounts in FIS-eBAM.

Note: The probability and impact will vary depending on the organization/business unit that is being audited.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Cash Management

Objective:

- Are there adequate controls to protect IBM cash assets?
- Is there an effective separation of duties among employees responsible for cash movements, bank administration, bank reconciliation, journal entries, and control of checks?
- Are there adequate controls for opening, closing, and maintaining bank accounts?
- Does management review periodically with each bank the financial terms and conditions applied to IBM?
- Are there adequate controls over bank fees/charges and interest calculations?
- Are bank account reconciliations performed on a timely basis and aged items resolved or addressed?
- Do the recorded cash receipts and disbursements represent aggregate collections and payments during the period? Are they properly applied to the appropriate ledger accounts?
- Is there an ongoing cash forecasting process in place for identifying cash requirements and availability?
- Are there adequate documented policies and procedures, including identified process risks.
- Is there adequate documentation and timely management review of established measurements.

Risk/Exposure:

- Failure to maintain proper control in cash activities could result in asset loss.
- Inadequate separation of duties allows improper movements of cash which could be concealed in bank reconciliations; blank check stock could be used to make improper disbursement which could be disguised in reconciliations.
- Undocumented bank relations, bank accounts opened/closed without the required approvals or bank account not established in the ledger could result in improper activities and unauthorized or fraudulent transactions. Not reviewing purpose and/or benefit of each account on an ongoing basis could result in an excessive operational and financial cost due to bank relationships/accounts not needed.
- Not reviewing periodically financial terms and conditions with each bank according to financial market trend could result in additional operating and financial costs to IBM.
- Failure to maintain proper control over bank fees/charges and interest calculations could result in inaccurate fees, charges, or interest expense.
- Not reconciling bank accounts on a regular basis could result in a undetected asset loss caused by fraud or errors.
- Inadequate control for cash receipts and disbursements create opportunity for the fraudulent diversion of the company assets.
- Failure to forecast properly cash position could expose IBM to loss of interest income or excess borrowing costs.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Cash Management

Primary Test: Bank Relations (CM001.P)

Objective:

The purpose of this review is to ensure that bank relations is in compliance with FIN181, and any exceptional approvals are obtained at the appropriate level of management.

The main areas to be tested in bank relations are:

- Opening bank accounts
- Closing bank accounts
- Bank Signatories
- Bank Fees
- Bank Account Recertification

Approach:

Open and Close Bank Accounts

- Obtain a list of bank accounts that were opened and closed during the audit period. This listing should include the bank name, bank account details, purpose of account, and ledger account details.
- Ensure that the annual bank certification was performed, properly documented, and the FIS-electronic Bank Account Management (FIS-eBAM) was updated.
- Request a positive bank confirmation with some banks that IBM has relationships with. Reconcile that list with the bank accounts in FIS-eBAM.

Opening Bank Accounts

- Ensure that there was a valid business need and supporting documentation to open an account.
- Review the bank account contract or application, and review if that documentation is according to IBM standards or reviewed by Legal.
- Review if the approvals were obtained as FIN181.
- Bank accounts must be entered into the FIS-eBAM within 5 days after the bank has confirmed that the account has been opened.
- Bank accounts must be created in IBM's general ledger no later than 45 days from bank account opening and reconciled according to the WW reconciliation standards.
- Ensure that the bank provided written confirmation that the account was open.

Closing Bank Accounts

- Review the justification to close a bank account (ensure that there is no pending business or receivables with the account prior to closing).
- Ensure treasury received written bank confirmation that the account was closed, and the balance was zero.
- Ensure treasury communicated to accounting about the bank account closure and maintain the evidence as required by FIN181, including the last reconciliation or evidence that this reconciliation exists.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Cash Management

Primary Test: Bank Relations (CM001.P)

(Continued)

- Review FIS-eBAM to ensure the account was closed.

Bank Signatories

- Request a list of changes in authorized signatories during the audit period.
- Select a sample of banks and ensure that they were notified timely about the changes.

Bank Fees

- Treasury is responsible to perform an analysis of bank fees and ensure their charges are in agreement with the contract or fees' schedules.
- Verify that bank fees, charges, value dates and interest rates are defined in a contract or equivalent (e.g., prospectus, fee schedule).
- When applicable, verify that terms and conditions were formally reviewed with each bank on a regular basis.
- The bank fees analysis must be reviewed and approved by treasury management.
- Based on the financial terms and conditions with each bank, select a sample of accounts/transactions to determine whether:
 - Bank fees/charges were applied to IBM in accordance with the contract.
 - Calculation of interest on bank overdraft/balance was verified and interest were applied to IBM according to the contract.
 - Fees, charges, and interests were recorded accurately in the ledger.

Data Protection

- As part of the bank relations analysis, is important to validate if the appropriate country or regional data protection requirements were properly established, being performed, and monitored by Treasury operations.
- Usually, the bank environment is significantly regulated about data protection and non-disclosure terms and conditions. However, an overview of the procedures related to personal information (PI) or sensitive personal information (SPI), including but not limited to the General Data Protection Regulation (GDPR) or any other similar, should be validated.
- Additionally, confirm all types of activities or transactions where the anonymization or pseudonymization information must be provided due to those requirements.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Cash Management

Primary Test: Cash Flow Forecasting (CM002.P)

Objective:

The purpose of this review is to ensure treasury puts together an accurate cash forecast to avoid bank overdrafts or short-term loan fees in the event of being short of cash. Similarly, holding excess cash would result in un-invested capital and missing opportunities.

Approach:

Select a sample of cash forecast files, including measurement and management reviews and validate:

- Review cash flow forecasting process and verify that cash availability and operations cash flow requirements are clearly identified and documented.
- Compare projection with actual cash flow results to confirm validity of the process.
- Determine how often cash flow forecasts were being performed and what were the assumption parameters that went into the forecast.
- Was there independence by the individual responsible for preparing forecasts to the individual responsible for execution of trade activities.
- Review management identified process measurements, reviews, and difference's analysis between estimate and actual results.
- Review the daily cash forecast and ensure that previous day closing balance matches with next day opening balance.
- Review if there are any forecast benchmarks (for variances) set by the Regional Treasurer or other documented evidence of how it was defined.
- Ensure that treasury is in communication with the appropriate organizations to obtain input for cash forecast and major discrepancies were being addressed. Relevant organizations would include accounts payable, accounts receivable, IBM Financing, tax, etc.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Cash Management

Secondary Test: Wire Transfer (CM003.S)

Objective:

To ensure that electronic funds and wire transfers are properly authorized and sent/received to/from approved accounts.

Approach:

Review the wire transfers process and disbursement of funds and determine whether:

- Adequate supporting documentation exists or was received and reviewed by treasury personnel prior to release the funds
- Adequate documentation existed in files that confirms required approvals and other process requirements.
- Electronic fund transfers were properly approved by management.
- The transaction was signed by an authorized person.
- The bank transfer confirmation was received and stored by IBM.
- The electronic funds/wire transfers were recorded in the general ledger, and they were reconciled against the bank statements.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Cash Management

Primary Test: Bank Reconciliations (CM004.P)

Objective:

To ensure that cash accounts are being reconciled with bank statements on a timely and accurate basis, and according to accounting procedures. Also, ensure auxiliary ledger reports were in line with summary ledger balances and that auxiliary records include reconciling items from the bank reconciliations.

Approach:

Review bank reconciliations for all accounts for three months preceding the announcement of the audit. From these reconciliations, based on an adequate or risk-based sample, determine whether:

- Unreconciled items were tracked, and action plans were adequately documented.
- Reconciliations were completed by the close of the following month and any delay was properly documented or reported.
- Reconciliations were signed/dated by the analyst and reviewed/approved by appropriate management.
- Reconciliations were cross footed, totaled, and debits and credits agreed to bank statements and the general ledger.
- Auxiliary balances for all bank accounts were equal to the summary ledger accounts.
- Auxiliary details were aligned with bank accounts reconciliation, were proper aged and have actions plans for resolution.
- Review all cash in transit account analyses for the same period and determine if they were properly prepared and documented.
- Review the supporting documentation used to reconciled cash accounts and ensure that it is in accordance to the WW balance sheet reconciliation standards.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Cash Management

Secondary Test: Cash Funds and Blank Checks (CM005.S)

Objective:

To ensure controls in place for blank checks are in compliance with financial instructions. Also, ensure that the access to liquid assets (such as cash and blank checks) is restricted to authorized individuals.

Approach:

- Review the process of receipt, storage, and issuance of checks to compare with was recorded in the control book and the last monthly physical inventory.
- Review the blank check inventory and ensure management has review it for accuracy.
- Review the process to discard blank checks.
- Perform a surprise blank checks and cash fund (at least one) count, preferably on the first day of the audit, to determine if all cash and blank checks are accounted for. Review control over these assets and ensure regular counts were being performed.
- Review disbursements for the prior two months for the same cash fund/selected checks to determine if disbursements were properly supported by original receipts, they were approved for legitimate business purpose, and they were properly recorded in the ledger.
- Determine who has access to cash, cash receipts, and blank checks for SOD concerns.
- If available, request a data pull from security with all the employees who badged into the safe room and reconcile against the list with authorized employees.
- If available, request the security video for employees accessing the safe room and reconcile against the safe room sign in log for accuracy.
- Ensure that dual access is needed (i.e., one employee has the safe key and another one has the password) in order to open the safe.
- Ask the client what efforts have been made to minimize the use of checks.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Cash Management

Secondary Test: Cash Receipts (CM006.S)

Objective:

To ensure that all cash receipts are properly safeguarded, promptly deposited, properly applied, and are recorded timely and accurately in the ledger.

Approach:

Document the cash receipts process and determine whether:

- Funds received are supported by receipt.
- Checks are logged and restrictively endorsed upon receipt.
- Checks received in the mail were being opened from a central location in the presence of at least two individuals.
- Receipts are adequately secured until deposit.
- Receipts are deposited in the bank on a daily basis.
- Receipts are recorded in the ledger in a timely manner, and they are applied to the correct customer account as the support documentation.
- Receipts, which could not be immediately applied to customer accounts, are properly followed-up and promptly resolved.
- Reconciliations of receipt logs to cash applied/bank deposit are performed regularly.
- Controls over transfers between customer accounts are adequate to prevent/detect lapping and netting between customer accounts.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 1. Cash Management

Secondary Test: Cash Transactions (CM007.S)

Objective:

To ensure all cash transactions were for valid business reasons, properly recorded in the ledger and in the bank statements and had required approvals.

Approach:

From a sample of bank accounts select a sample of transactions (receipt and disbursement of funds) to determine whether:

- Transactions were for valid business reasons.
- Transactions were properly executed and included necessary supporting documentation.
- Transactions were recorded in the bank statement and in the general ledger.
- Transactions were recorded upon required approvals.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Investment Portfolio

Objective:

- Is there adequate separation of duties regarding investment portfolio transactions?
- Is liquidity managed through cash flow forecasting for timing the maturities of short-term investments?
- Are transactions properly analyzed by treasury department to ensure the best return on IBM assets?
- Have broker/dealer relationships been authorized by management for all investments including derivatives?
- Are all transactions within the delegated authorizations, properly approved and in compliance with corporate and investment policies?
- Are confirmations required for transactions sent by IBM to the banks, brokers, and dealers (and acknowledged by banks, broker and dealers)? Were these confirmations reconciled to transaction record?
- Are all securities properly safeguarded by use of a custody account?
- Are all investment transactions properly recorded and reconciled by accounting on timely basis?
- Are interest calculations accurate and properly recorded in the ledger?
- Are controls and procedures in place for investing in derivatives instruments?
- Is there adequate documented policies and procedures, including identified process risks?
- Is there adequate documentation of timely management review of established measurements?

Risk/Exposure:

- An inadequate separation of duties could result in an inability to detect unauthorized transactions and could lead to a significant loss of assets.
- Failure to manage liquidity through cash flow forecasting could results in missed investment opportunity or low yielding short-term investments.
- Failure to properly record transactions could result in an inability by management to detect unauthorized investments or counterparty, low yielding, or high-risk investments.
- Failure to obtain management authorization for investment transactions could results in a loss of assets by investing in inappropriate or high-risk securities.
- Failure to reconcile investment transactions could result in an inability to detect transaction execution error, unauthorized transactions, or loss of assets.
- Failure to properly safeguard securities could results in loss of assets.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Investment Portfolio

(Continued)

- Failure to properly account for all transactions could results in inaccurate financial statement.
- Failure to review interest calculations or interests not properly recorded in the ledger could result in financial loss to IBM and inaccurate financial statement.
- Failure to comply with corporate directives regarding the types of investments could result in a significant loss of assets and financial statements impact.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Investment Portfolio

Primary Test: Valuation and Control of Portfolio (IP001.P)

Objective:

To ensure investments are properly reviewed by management and in line with treasury policies. In addition, ensure transactions are properly approved/recorded/reconciled, that securities are properly safeguarded, and if all hedging positions are valued consistently and at market prices.

Approach:

- Verify that treasury control reports for investment maturity and portfolio were reported to management on a timely basis.
- Reconcile treasury control reports for investments to the file of banks/brokers involved in investment instruments issuance and ensure that all investments were included/reflected by a relevant category in both treasury and accounting records.
- Request all reconciliations of ledger investment accounts for the month prior to the audit to determine if they were prepared and approved on a timely basis.
- Review the treasury operation's investment strategies, policies, and delegation letters from to determine the following:
 - Whether current delegations of authority were available for review and did these delegations provide specific responsibility and authority in the following areas:
 - Trading
 - Recording of transactions
 - Cash movements (settlement)
 - Valuations
 - Management reporting
 - Financial reporting
 - Risk management
- Were corporate guidelines clearly communicated to the regions, and were they adhered to.
- Were there any deviations from the corporate mandated hedging program risks and exposures. If yes, are the deviations properly identified and approved by the IBM Treasurer and specified in the regional guidelines.
- Were there defined processes for hiring and training personnel, with specific guidelines for meeting minimum education or licensing requirements?
- Was there evidence of ongoing review by management and traders of current policies, trader limits (dollars and transaction type) at least annually.
- Was the valuation determined for each individual position, as well as the entire portfolio.
- Was the valuation determined on a consistent basis (daily, weekly, monthly, quarterly, etc.), and as required by the corporate instructions?

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Investment Portfolio

Primary Test: Valuation and Control of Portfolio (IP001.P)

(Continued)

- Was the current portfolio valuation compared to the previous valuation performed? Did the comparison include an explanation for the changes in the number of positions and the dollar value from one period to the next?
- Have the valuation model's assumptions been validated?
- Does an external vendor perform the mark-to-market valuation, or does IBM perform this valuation internally?
- If an external vendor performs the valuation:
 - Was management aware of the assumptions used in the model?
 - Were the assumptions reasonable when compared to historical performance?
 - Were the assumptions updated (e.g., daily, or periodically)?
 - Were the results compared to results obtained from another reputable external vendor, at least periodically?
 - If there was a comparison, were any differences identified? If so, were these differences reconciled, documented, reported to management, and appropriate action taken to address these differences.
- If the mark-to-market valuation was performed internally:
 - Was management aware of the assumptions used in the model and were the assumptions reasonable when compared to historical performance?
 - Was an external vendor engaged to independently value the portfolio for comparison to internally generated results?
 - If there was a comparison, were any differences identified? If so, were these differences reconciled, documented, reported to management, and appropriate action taken to address these differences.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Investment Portfolio

Primary Test: Cash and Investment Planning (IP002.P)

Objective:

To confirm that excess cash is placed in short-term investments to maximize interest income.

Approach:

Local cash balances must be kept at minimum. Any excess cash, if possible, must be daily remitted to Dublin Treasury Center (DTC) for aggregation and investment. In case that there are impediments to send excess cash to DTC, the countries should evaluate alternate types, considering three objectives when determining an appropriate investment:

- **Safety:** The financial condition of the obligor must be among the most stable investments available within the local marketplace.
- **Liquidity:** When applicable, instruments must have a ready secondary market that would allow the transaction to be readily turned into cash or otherwise terminated at current market value.
- **Yield:** Safety and liquidity are the primary investment objectives. Yield and return should be considered once safety and liquidity guidelines have been met.

Cash Investments

- Review the cash investments and ensure they are in accordance with the investment and counterparty limit guidelines.
- Review if the investments are in functional currency.
- Review the investments portfolio as the country investment grade sovereign ratings as defined by Moody's Investors Service. Are the investments in government securities or bank liabilities accordingly to the maximum duration allowed?

Other Criteria

- Verify if short and long-term forecasts were used for maturity planning and investment strategy purpose.
- Verify if a management process decision exists to invest excess cash in a timely manner.
- Review cash flow forecasting process and verify that cash availability and requirements are identified.
- Determine how often cash flow forecasts were being performed and what were the assumption parameters that went into the forecast.
- Was there independence by the individual responsible for forecast to the individual responsible for execution of trade activities.
- Compare projection with actual cash flow to confirm validity of the process.
- Review management identified process measurements and compare results to what was forecasted. Analyze differences between estimate and actual results.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Investment Portfolio

Primary Test: Broker/Dealer Relationships (IP003.P)

Objective:

Determine if there is an ongoing process to validate and assess the credit worthiness of new and existing broker/dealer to ensure that all transactions were in compliance with investment policies and authorized by management.

Approach:

- Review the approved list of authorized IBM traders and ensure that banks, brokers and dealers had copies of the approved list.
- Ensure trades were independently reviewed, properly authorized, and in compliance with the investment policy and applicable corporate instructions.
- Select a sample of recent additions to the broker/dealer list and determine:
 - Were counterparty's selected in accordance with established risk guidelines?
 - Determine if all related corporation's activities were considered for purposes of establishing counter party limits.
 - Determine if reputation was adverse (recent negative publicity, history of litigation), if so how was negative publicity dealt with.
 - Were agreements executed prior to business being conducted with the broker/dealer?
 - Were the decisions to add counterparties reviewed and approved at the appropriate management level as the counterparty guidelines?
 - Was business activity sufficiently allocated amongst counterparties so that there was sufficient capacity to process additional transactions at all times without exceeding counter party limits?
- Review the broker/dealer listing and determine:
 - Was there a systematic process to periodically evaluate the broker/dealer and to update the listing.
 - Were limits set for each broker dealer.
 - Was worldwide aggregate activity (concentration) accumulated and monitored for each broker/dealer and all associated alliances.
 - If the counterparty's limit was exceeded, was there a process to handle the excess, including with the appropriate approvals or exceptions.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 2. Investment Portfolio

Primary Test: Interest Income (IP004.P)

Objective:

To ensure that interest and bonuses on deposit/securities are accurate in accordance with contractual terms and conditions. Also, ensure interest income is properly recorded in the ledger.

Approach:

Review control reports for interest bearing deposits/securities to determine whether:

- Interest receivables were reflected in the cash flow forecasting.
- Interests and bonuses were accurate in accordance with the contractual terms and conditions.
- Interests income were properly recorded in the ledger.
- For country treasury operations using the Dublin Treasury Center for overnight investments, ensure that interest receivables were accurately calculated/received for funds on deposit with their call account.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 3. Borrowing Management

Objective:

- Is there an adequate separation of duties regarding debt transactions?
- Is liquidity managed through cash flow forecasting for rolling over maturing loan or anticipating borrowing needs?
- Are debt transactions approved within delegated authorizations and in line with corporate policies?
- Are debt transactions properly reported to management? Are debts reviewed by treasury department to ensure the best interest rate?
- Are debt transactions properly recorded and reconciled by accounting on timely basis? Were interest expenses properly recorded?
- Are payments made promptly and accurately according to treasury strategy (e.g., excess cash) and contractual terms and conditions?
- Is there adequate documentation of policies and procedures, including identified process risks.
- Is there adequate documentation of timely management review of established measurements.

Risk/Exposure:

- An inadequate separation of duties could allow improper and unauthorized debt transactions which could lead to a significant loss of funds.
- Failure to manage liquidity through cash flow forecasting could result in unnecessary financial exposure and costs.
- Failure to obtain management authorization and approvals, and control debt transaction's portfolio could allow improper or unauthorized debt transactions, unnecessary financial exposure, and costs.
- Failure to properly account for debt transactions or interest expenses have an impact on the company's financial position. The failure to reconcile debt transactions could result in an undetected loss of funds or theft of debt instruments.
- Failure to make payments for debts not in line with contractual terms and conditions could result in additional financial cost.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 3. Borrowing Management

Primary Test: Valuation and Control of Debts (BM001.P)

Objective:

To ensure debt transactions are executed in accordance with delegated authorization and are properly recorded. Also, ensure debts are incurred through a competitive evaluation process.

Approach:

- Verify if treasury information control reports for borrowing maturities were reported to management or measured on timely basis.
- Reconcile control reports for debt schedule to the file of banks/brokers involved in debt issuance and ensure that all debts were included and reflected by a relevant category in both treasury and accounting records.
- If applicable, verify that all bank debt instruments are prenumbered and kept in locked storage under a perpetual inventory system. Review reconciliations at month-end to physical inventory.
- Determine whether policies and procedures were adequately documented.
- Select a sample of loan and other debt transaction (e.g., leases) activity and determine whether:
 - Debt transactions were documented and were executed with banks/brokers properly authorized.
 - Debt instruments were issued according to policies and procedures and amount approved within delegated authorization.
 - Debt transactions were justified for business needs, and business needs were properly documented.
 - Debts were valued to ensure the best interest rate, and debt valuation were properly documented.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 3. Borrowing Management

Primary Test: Cash and Debt Planning (BM002.P)

Objective:

To confirm that net cash position is planned to minimize interest expenses to the company by anticipating borrowing needs.

Approach:

- Ensure short and long-term forecasts were used for maturity planning and borrowing strategy purpose.
- Verify that the issuance of debts is consistent with cash forecasting and objectives.
- Review last three-month bank accounts level, prior the audit, and determine whether lines of credit/overdrafts were managed to minimize interest expenses.
- Review cash flow forecasting process and verify that cash availability and requirements were being identified.
- Compare projection with actual cash flow to confirm validity of the process. Additionally, check if a management review exists over that comparison.
- Determine how often cash flow forecasts were being performed and what were the assumption parameters that went into the forecast.
- Was there independence by the individual responsible for forecast to the individual responsible for execution of trade activities.
- Review management identified process measurements and compare results to what was forecasted. Are the differences analyzed or addressed for improvements?

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 3. Borrowing Management

Secondary Test: Payment of Debts (BM003.S)

Objective:

To ensure that debts are paid promptly and accurately, in accordance with terms and conditions agreement. Also, ensure payments are properly recorded and reconciled.

Approach:

Select a sample of short and long-term debts and determine whether:

- All required payments for debt were included in the cash flow forecasting.
- Payments were timely, accurate, and consistent with the contractual terms and conditions.
- Payments were properly recorded against to the specific debt. So, that update debt balance represented the outstanding value in line with the contractual due date.
- Payments were reconciled to the general ledger debt/bank.
- Debts outstanding were properly reconciled.
- Debt payment was clearly allocated between principal and interest.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 3. Borrowing Management

Primary Test: Interest Expenses on Debts (BM004.P)

Objective:

To ensure that interest payments on short and long-term debt are made in accordance with the requirements of debt instruments, and that they have been recorded correctly. Also, ensure interests are accrued for outstanding debt promptly and accurately.

Approach:

Review control reports for interest payments schedule on debts and verify the following:

- Interest payables were reflected in the cash flow forecasting.
- Payments were prompt/accurate in accordance with contractual terms and conditions.
- Interest expenses were properly recorded in the ledger.
- Interest accrual schedules were reviewed and reconciled to total debt outstanding, as of statement date, and calculations were accurately recorded in the ledger.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 4. Intercompany Transactions

Objective:

- Are controls in place to ensure that settlements of intercompany charges are made accurately and on a timely basis?
- Are controls in place to ensure that dividends and royalties are paid on a timely basis?

Risk/Exposure:

Delays in making intercompany settlements and dividend/royalty payments may expose the corporation to the loss of interest income, foreign exchange losses, and adverse tax treatment.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 4. Intercompany Transactions

Primary Test: Settlement of Intercompany Charges (IT001.P)

Objective:

To ensure settlement of intercompany charges, including dividends and royalties, are executed accurately, in accordance with management authorizations, IBM policies, and Treasury and Tax guidelines.

Approach:

- Ensure intercompany payment notification procedures are in place to provide accountability for payments to and receipts from IBM affiliate companies.
- Ensure that separation of duties was appropriate in the areas of authorizations and movement of funds for settlement of intercompany charges.
- Review local regulations regarding the netting of intercompany debts and ensure that country procedures comply with local regulations. In some countries, this practice is forbidden for tax, currency exchange, and legal reasons.
- Review settlements of charges for goods/services to determine whether:
 - The net monthly settlement balance due/receivable was properly supported by source documentation from Intercompany Accounting.
 - Payments were accurate, in agreement with payment proposals approved by Intercompany Accounting and CHQ Treasury, made on the due date, and with the proper approvals.
 - Transfers were made to the valid bank account established in the corporate database for intercompany payments.
 - Transfers outside the netting system were promptly advised to the receiving location and confirmation of receipts were provided according to CHQ guidelines.
 - Settlements made beyond the normal due date and/or in other than the currency of the country shipping the goods/services were approved within the delegated authority.
 - Dividend/Royalty payments were accurate, as the dates and amounts advised to CHQ, made on due date, and with the proper approvals.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 5. Foreign Exchange Management

Objective:

- Is there adequate separation of duties regarding foreign currency exchange transactions?
- Are foreign exchange transactions executed in accordance with corporate guidelines? Are transactions properly documented/approved?
- Are transactions reviewed by management to ensure the best foreign exchange rate, and, when applicable, with the competitive bid?
- Are controls and procedures in place for hedging currency risks and exposures?
- Are foreign assets and liabilities revaluated/devaluated in accordance with accounting instructions? Are exchange gains or losses calculated accurately and properly recorded in the ledger?
- Is there adequate documentation of policies and procedures, including identified process risks?
- Is there adequate documentation of timely management review of established measurements?

Risk/Exposure:

- An inadequate separation of duties allows improper/unauthorized or fraudulent exchange transactions which could lead to a significant loss of funds.
- Failure to obtain management authorizations or foreign exchange transactions not executed in line with corporate guidelines could result in currency exchange losses to IBM.
- Failure to review foreign exchange transactions could result in unnecessary costs and currency exposures.
- Failure to comply with corporate directives regarding hedging instruments or currency risk not properly hedged could result in a significant loss of company assets and negative impact to the financial statement.
- Failure to reevaluate/devalue all assets and liabilities in foreign currency or exchange gains/losses not properly recorded could result in inaccurate financial statement.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 5. Foreign Exchange Management

Primary Test: Foreign Exchange Transactions (FX001.P)

Objective:

To ensure that the needs to exchange currencies are clearly identified so that IBM can optimize their foreign exchange transactions, costs, and exposure. Also, ensure that foreign exchange transactions are executed within delegated authorization, according to the corporate guidelines, and were properly recorded in the ledger.

Approach:

- Ensure that an adequate process is in place for managing foreign exchange operations based on the cash, guidelines, and competitive bidding requirements.
- Review the documentation utilized by the department and verify that all transactions are renumbered and accounted for in both treasury and accounting records.
- Based on the process established to manage foreign exchange operations, select a sample of transactions (e.g., high value of foreign-banking transactions, foreign payments/receipts) and determine whether:
 - Transactions were properly reviewed by management.
 - Transactions were properly documented (from the request to the competitive bid and contracts) and approved.
 - Transactions were executed within the delegated authorizations and in accordance with the guidelines.
 - Exchange rates/currency conversion were accurate and properly registered, also for accounting purposes.
 - Transactions exchange gains or losses were properly calculated and recorded in the ledger.
- Review correspondence files between IBM and local regulatory bodies, national regulatory bodies, brokers, banks, customers, counterparties, and any other correspondence, contracts or regulatory reports.
- Notes and letters should be investigated for any signs of violations of trading regulations, and complaints.
- Determine whether there were any tax or legal penalties assessed related to ongoing hedging activities.
- Has the cause of the penalty been investigated, reported to appropriate levels of management, and were appropriate actions taken to prevent reoccurrence?
- Were penalties properly accounted for?
- Select a sample of reports (obtained from auditee and CHQ) and test for:
 - Detailed transactions reported by country, trader, and type of transaction.
 - Market valuations are compared to the previous day, and any significant price or portfolio discrepancies are investigated.
 - Review of foreign currency cash flows.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 5. Foreign Exchange Management

Primary Test: Currency Exposures (FX002.P)

Objective:

Ensure that currency exposures hedging is made within the delegated authorizations and according to corporate guidelines.

Approach:

- Ensure that an adequate process is in place for identifying and hedging currency exposures.
- Ensure that any specific hedging instruments program at country level such as currency swaps or forward rate agreements was properly approved by CHQ, TC, or as required by the guidelines.
- Review the hedge activity against the exposure analysis and make sure that was properly controlled, monthly valuated (MTM), and recorded for accounting purposes.
- Review the assets and liabilities in foreign currency and determine whether:
 - Currency exposures were identified, and hedging was agreed with CHQ/TC.
 - Hedging instruments were contracted with the proper approvals, and they were in compliance with country and local regulation.
 - Any natural hedges (offsetting of equal and opposite currency risks) were properly approved.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 5. Foreign Exchange Management

Primary Test: Accounting Controls (FX003.P)

Objective:

Ensure that accounting controls over assets and liabilities in foreign currencies are properly accounted.

Approach:

Select a sample of high-value foreign assets and liabilities and determine whether:

- Assets and liabilities were revaluated/devaluated according to accounting instructions.
- Exchange rates were accurate.
- Exchange gains or losses due to revaluation/devaluation were properly calculated and recorded in the ledger.
- Accounts were reconciled in both local and foreign currency.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 6. Investment in Derivatives

Objective:

Determine if all derivative positions were properly authorized and approved, evaluated consistently, in compliance with the risk management policy, and were accounted in accordance with Financial Accounting Standard (FAS) 133.

Risk/Exposure:

The lack of adherence to corporate treasury and risk management policies and strategies, and incomplete reporting of derivative activity, could result in unauthorized trading activity, inaccurate management and financial reports, and financial losses to IBM due to inadequate currency exposure.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 6. Investment in Derivatives

Primary Test: Investment in Derivatives (ID001.P)

Objective:

To determine whether the usage of risk management instruments undertaken by local units adheres to the worldwide treasury objectives and the risk management policy statement for the use of risk management instruments (RMI).

Approach:

Select a sample (or all if reasonable) of derivative transactions from the complete universe of RMIs and ensure the following:

- All derivative transactions have prior approval by the Assistant Treasurer or as specified in the risk management policy statement for the use of RMI.
- Were there any transactions that appeared speculative?
- All derivatives transactions were approved investments and abide by the safety, liquid, and yield objectives. Verify that any exception to the corporate guidelines received prior approval of the IBM Treasurer.
- The trade was executed in accordance with the planning requirement for that particular program as outlined by Risk Management.
- The correct underlying transaction was linked to the correct hedge per FAS133 requirements.
- Do the derivative transactions agree with the hedging program guidelines? The derivative transactions were adequately documented, timely recorded, and accurate (i.e., were trade tickets and blotters prepared in a timely and complete manner? Did they contain all necessary trade details?)?
- All trade confirmations for each transaction were sent directly to the local country's accounting department or settlements department, and details of the transaction were entered accurately into the ledger system. If a system was not utilized, ensure that a process exists to accurately summarize all transactions and report results to treasury management.
- Where applicable, review the underlying documentation of each transaction including trade confirmations and ISDA master agreements. Determine whether the terms and conditions of these documents were adhered to, including netting between the paying and receiving parties. Were contractual terms of transactions confirmed prior to settlement?
- Review management identified process measurements, where applicable check for inclusion of measurements for the following:
 - Hedge effectiveness (i.e., negative correlation to underlying risk)
 - Percent hedged
 - Costs of options
 - Total gains and losses
 - Underlying risks

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 6. Investment in Derivatives

Primary Test: Investment in Derivatives (ID001.P)

(Continued)

- Controls were in place to limit the risk of unauthorized transactions (i.e., requirements that traders obtain control numbers before executing transactions, and confirmations sent directly to the back office)?
- Determine whether there were any tax or legal penalties assessed related to ongoing hedging activities. Has the cause of the penalty been investigated, reported to appropriate levels of management, and were appropriate actions taken to prevent reoccurrence? Were penalties properly accounted for?

Currency of Investments

- If the functional currency is the local currency, verify that all investments were made in the local currency. If any exceptions exist, verify that authorization was previously received.
- If the functional currency of a non-US entity is not the local currency, verify that investments were made in either the local or functional currency so long as the local economy is not considered hyper-inflationary (as defined by FAS52) and the investment in nonfunctional currency has been approved by the Regional Treasurer or as required by the guidelines. If the local currency is hyper-inflationary, verify that local currency investments were hedged when economically practical and approved in advance by the Regional Treasurer or as required by the guidelines.
- In the cases when the functional currency is neither the local currency nor the US\$, verify that local investment policy was reviewed at least semi-annually with the Regional Treasurer or as required by the guidelines.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 6. Investment in Derivatives

Primary Test: Valuation of Derivatives (ID002.P)

Objective:

Determine if all derivative positions were properly authorized and approved, evaluated consistently, in compliance with the risk management policy. In addition, to determine whether material trends, and fluctuations in derivative positions were identified and investigated.

Approach:

Review the mark to market process for determining portfolio value, and obtain the most recent valuation of the portfolio to determine the following:

- Were all portfolio positions submitted for valuation individually by product type?
- Was the valuation determined periodically and on a consistent basis as required by corporate policy (i.e., daily, weekly, monthly, quarterly, etc.)? Were variations in portfolio performance clearly identified and explained?
- Have the derivative valuation model(s) assumptions been validated to ensure integrity?
- Does an external vendor perform the mark-to-market portfolio valuation to validate positions? If and external vendor performs an independent valuation, ensure that the valuation is all inclusive (i.e., valuation formulas updated to reflect current market events).
- If there were any discrepancies identified, were these differences reconciled, documented, reported to management, and appropriate action taken to address these differences?
- Has management implemented a process to anticipate and react accordingly to significant day to day market movements and its effect on the portfolio (i.e., stress testing for interest rates movements and the impact on the portfolio as, for example, what would be the impact if the market moves up or down by 100, 200, and 300 basis points)? If yes, test for the following:
 - Were these tests conducted periodically (i.e., weekly) and results reviewed and approved by treasury management?
 - Does the testing consider market correlation (e.g., under significant market movement assumptions, the markets may all move in the same direction)?
 - Is the portfolio monitored for opportunities to unwind (liquidate) derivative positions? How was the information obtained?

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 6. Investment in Derivatives

Primary Test: Financial Accounting Standard 133 (ID003.P)

Objective:

To determine if IBM risk management instruments are properly evaluated to comply with Financial Accounting Standard (FAS) 133 (accounting for derivatives instruments and hedging activities).

Approach:

- Verify all individual transactions are captured in accordance with the specific requirements of FAS133.
- Verify that the key FAS133 reports as generated by the local system, or method for generating reports, serve as a reasonable basis for the booking of accounting entries.
- Verify that the accounting treatment being applied by the system functionalities for all new and existing derivative instruments (both hedges and non-hedges) is in accordance with the detailed requirements of FAS133.
- Verify that the interface between the treasury and accounting systems facilitates the booking of entries in accordance with the detailed requirements of FAS133.
- Verify that the accounting system ultimately produces accounting entries consistent with FAS133 compliance requirements.
- Verify that system enhancements have not impacted the ability to report results under existing reporting requirements.
- Verify that all information required for FAS133 disclosure reporting can be retrieved from the system or general ledger.

TREASURY OPERATIONS AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 7. Policies and Procedures

Objective:

Were policies and procedures established to ensure that trading activities of the treasury group were consistent with current corporate strategy? Did trade activity adhere to policy limits and objectives established in the written procedures?

Risk/Exposure:

The failure to establish appropriate policies could result in unauthorized investment activity, failure to meet corporate objectives, and financial loss.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 7. Policies and Procedures

Corporate and Regional Policies and Procedures (PP001.P)

Objective:

To determine whether there were documented policies and procedures at both the corporate and regional levels, that defines the treasury investment and borrowing strategies, the risk management hedging strategy, and the approved listing of derivative instruments. In addition, did the policies and procedures provide sufficient detail to guide the program.

Approach:

Review the corporate and regional hedging strategies, and delegation letters to determine the following:

Corporate

- Were hedging program risks, exposures, and limits properly identified and clearly defined in written policy documentation? Were contingency plans for all risks in excess of limits defined in the written policy documentation (i.e., counterparty settlement risk in excess of limits)?
- Policies should identify exposures relevant to the following risks:
 - Market risk
 - Market liquidity risk
 - Basis or Correlation risk
 - Credit risk
 - Legal risk
 - Settlement risk, including time zone considerations, and transactions where the notional amount is exchanged on the day of settlement (i.e., cross-currency swaps)
 - Funding liquidity risk
 - Operational risk
- Limits should be properly identified relevant to:
 - Trader limits
 - Counterparty position limits
 - Levels of unhedged market exposure
 - Stop-loss limits
 - Open positions by derivative type, showing notional amounts, maturity dates, and items hedged
 - Margin limits
- Were measurement reporting requirements established to properly identify, monitor, and track risks and risk limits against policy objectives and strategies?
- Were there defined processes for hiring, and training personnel, with specific guidelines for meeting minimum education or licensing requirements?
- Was there a documented process for monitoring ongoing global events which might impact the hedge program at corporate and regional levels? If so, were outputs or warnings, resulting from this process, clearly and timely communicated to the regional levels?

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 7. Policies and Procedures

Corporate and Regional Policies and Procedures (PP001.P)

(Continued)

- Were guidelines sufficient for identifying exceptions to policy, including regional variations? Were there procedures for documenting and resolving these exceptions?
- Was there evidence of ongoing review by management and traders of current policies, trader limits (dollars and transaction type), and delegation capabilities at least annually? For any changes, was there adequate documentation to demonstrate appropriate approvals?

Regional

- Were corporate risk management strategies and guidelines clearly communicated to the regions, and were they adhered to? Were there any deviations from the corporate mandated hedging program? If yes, were the deviations properly identified and approved by the IBM Treasurer, and specified in the regional guidelines?
- Was there a documented process to identify transactions which appeared speculative (i.e., derivative activities are considered speculative if they increased risk, if their use has no relation to objectives specified by the policy, or if their use is not intended and expected to reduce business risks that have been identified)? If yes, was a documented exception to policy obtained, and approved by the IBM Corporate Treasurer?
- Were appropriate instruments and the risk management policy, documented in local procedures.
- Where applicable, review the underlying documentation of the transaction including trade confirmations and ISDA master agreements. Determine whether the terms and conditions of these documents were consistent and adhered to.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 8. Management Reporting

Objective:

To determine whether material trends, and fluctuations in investments or positions were identified and investigated.

In addition, determine if financial and transactional reporting is supported by the general ledger, and reported to IBM management and external bodies on a timely basis and accurate.

Finally, to determine whether appropriate and timely corrective action had been taken to resolve unusual situations, or exceptions which have been communicated to management by outside parties.

Risk/Exposure:

Failure to review trends and performance against plan or strategy could allow account(s) to contain erroneous or improper information, thereby distorting results and/or concealing irregularities.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 8. Management Reporting

Primary Test: Management Reporting and Review (MR001.P)

Objective:

To determine whether material trends, and fluctuations in investments or positions were identified and investigated. In addition, determine if financial and transactional reporting is supported by the general ledger, and reported to IBM management and external bodies on a timely basis and accurate.

Approach:

- Review management identified process measurements, check for inclusion of measurements for the following:
 - Hedge effectiveness (i.e., negative correlation to underlying risk)
 - Percent hedged
 - Costs of options
 - Total gains and losses
 - Underlying risks
- Select a sample of reports (obtained from auditee and CHQ) and test for:
 - Detailed transactions reported by country, type of investment, and trader.
 - Market valuations are compared to the previous day, and any significant price or portfolio discrepancies are investigated.
 - Review of foreign currency cash flows.
 - Reports are reviewed daily by traders and their first line management.
 - A weekly review of trading positions by senior management is conducted.
 - Reports identify exception conditions (i.e., excessive losses, significant price, and portfolio valuation discrepancies, trader transactions beyond authorized limits).
- Were the entity's activities and the related balance sheet and off-balance-sheet instruments identified, and were risk management reports prepared?
- Concerning valuation, determine if:
 - Was the portfolio marked to market daily? Were market values taken after the close of trading? How were after close events handled?
 - Were controls adequate to ensure that all transactions were included in the reports to management? For transactions which cannot be priced, were other means used to value the transactions (i.e., impute price based on other securities, determining the cause for an absence of price).
 - Is the priced portfolio reconciled to the ledger (based on cost) to ensure all transactions are included?
- When applicable, does the hedging activities get properly reported to the Securities Exchange Commission (SEC)? Was the information, submitted for inclusion in SEC filings accurate (i.e., 10K, and annual report to shareholders)?

TREASURY OPERATIONS AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 9. Counterparty Risk Evaluation

Objective:

Review the counterparty limit (CPL) process and ensure the treasury operations are being monitored and reported monthly as required by the guidelines.

Risk/Exposure:

Inaccurate monitoring and reporting of CPL, including trading with unapproved counterparties, could lead to financial exposures going undetected and asset loss.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 9. Counterparty Risk Evaluation

Primary Test: Counterparty Risk (CR001.P)

Objective:

Determine if there is an ongoing process to monitor and report the daily activities and validate the credit worthiness of new and existing counterparties.

Approach:

- Select a list of all banks that IBM has relationships with during the audit period and request all limits assigned to them, including the changes.
- Were counterparty's approved and selected in accordance with established risk guidelines?
- Request a list of all changes and ensure the country treasury team monitor any changes in limits as required by the guidelines.
- Get a list of all accounts with all banks and ensure that the calculation of daily cash balances include all bank accounts.
- Was the counterparty assessment done prior to any trades being executed with that party? Usually, this process is not documented due to the operations procedures. If available, check the consolidated report of breaks to see if some breaks happened due to a treasury activity and validate the root cause and limit management process.
- In the event that IBM started a new relationship with a bank, was Risk Management informed so that the bank is assigned a limit?
- Was business activity sufficiently allocated amongst counter parties so that there was sufficient capacity to process additional transactions at all times without exceeding counterparty limits?
- Get a list of all subsidiaries and joint ventures and ensure that they are performing counterparty limit monitoring.
- Get a list of all counterparty limit breaks and review the reason for the break. If they are recurring breaks, request an action plan from the treasury team to avoid breakages in the future.
- Review the process to retrieve daily cash information and assess the adequacy. For example, some countries have very manual processes (i.e., call the bank to request balance and write it down on a piece of paper).

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 9. Counterparty Risk Evaluation

Primary Test: Counterparty Risk (CR001.P)

(Continued)

- Review the counterparty listing and determine:
 - Is there a systematic process to periodically evaluate the counterparties, and to update the counterparty listing?
 - Was there evidence that all were being reviewed regularly? An excessive number of counterparties could indicate cursory or incomplete review. An insufficient number could result in an inability to conduct certain trades.
 - Was there a process to monitor ongoing events (i.e., credit downgrades)?
 - Was there a process to communicate such events to traders in a timely manner?
 - As a result of any identified unfavorable events, did management update the counterparty profile on the system?
 - Was an analysis of positions with the counterparty reviewed, and appropriate actions taken regarding those positions?
 - Were limits set for each counterparty or for certain industries? Was worldwide aggregate activity (concentration) accumulated and monitored for each counterparty, and all associated alliances? If the counterparty's limit was exceeded, was there a process to handle the excesses?

TREASURY OPERATIONS AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 10. Disbursements

Objective:

Are all expenses claimed reasonable, documented, approved, and in compliance with IBM policies? Do any expenses claimed using Travel Expense Accounts (TEAs), Miscellaneous Expense Accounts (MEAs), Electronic Check Request (ECR), Purchase Order Exempts (PO Exempt - EMEA only), and Invoice Only Invoices (IOI - AP only) represent a bypass of the procurement process?

Risk/Exposure:

Improper or invalid payments may be made to employees which could result in inappropriate or excessive expenses being incurred by IBM.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 10. Disbursements

Primary Test: Procurement Bypass (DS001.P)

Objective:

To determine if controls are in place to prevent the bypass of procurement in selecting suppliers, making purchases, and/or disbursements.

Approach:

- Review documentation and management policies to validate the control measures listed below. In selecting a sample, refer to the Global Accounts Payable website. Additionally,
 - in North America refer to the Electronic Check Request (ECR) approved categories.
 - in EMEA refer to the PO Exempt procedures.
 - in AP refer to Invoice Only Invoices (IOI) procedures.
- Any expenses that do not fall into these categories must go through Procurement.

Note: Some categories also specify monetary limits which may assist in selecting a sample.

- Review ECRs, PO Exempts, or IOIs from the previous year. If the population is too large, select a sample. Repeated requests for payment to the same supplier and/or from the same requestor could be a sample selection criterion. Verify that expensed items did not require procurement involvement.
- Review a selection of the travel expense accounts (TEAs) and miscellaneous expense accounts (MEAs) to verify that expensed items did not require procurement involvement.
- Determine if management had procedures in place to prevent the bypass of procurement. If a bypass did occur, validate that control procedures were implemented in a timely manner to ensure that repeat bypasses would be prevented. If applicable, review any control risk assessment (CRA) measurements that the organization may have in place for procurement bypasses.
- Obtain a listing of suppliers used in support of the organization for general purchasing and select a sample of suppliers to confirm that they had been approved by procurement and that fair value analysis had been performed.

TREASURY OPERATIONS AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 11. Acquisition Integration

Objective:

Review the acquisition integration process to ensure IBM takes possession over all the bank accounts of the acquired entity.

Risk/Exposure:

Inadequate management of acquired entity banking relationships and access to the bank accounts could result in unauthorized activities and financial loss.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 11. Acquisition Integration

Primary Test: Acquisition Integration (AI001.P)

Objective:

Review the acquisition integration process to ensure IBM takes possession over all the bank accounts of the acquired entity.

Approach:

Request a list of all acquisitions during the audit period:

- Within 30 days, treasury is responsible to send a letter to all known banks and request a positive confirmation of all accounts.
- Ensure that counterparty limits were assigned up to 10 working days post the integration.
- Review the authorized signatories and ensure that IBM finance managers were added as authorized signatories.
- Request evidence that IBM took possession of all manual payments and bank seals.
- Ensure that the corporate development team maintains communication with treasury related to acquired companies.
- Ensure that all accounts were added to the FIS-eBAM and IBM ledger account as required by FIN181.

TREASURY OPERATIONS AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 12. Share Repurchases

Objective:

Review the share repurchases process to validate that share repurchases have been properly authorized and are executed in accordance with the Securities and Exchange Commission (SEC) rules.

Risk/Exposure:

Unauthorized share repurchase transactions could lead to financial loss and negative consequences for the company's strategic plan. Share repurchases that are not executed under the Securities and Exchange Commission (SEC) rules could lead to litigation, fines, and damage IBM's reputation.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 12. Share Repurchases

Primary Test: Share Repurchases (SR001.P)

Objective:

Review the share repurchases process to validate that share repurchases have been properly authorized and are executed in accordance with the Securities and Exchange Commission (SEC) rules.

Approach:

Request a universe of all shares repurchase transactions performed during the audit period. Review the transactions and validate that they were appropriately transacted based on:

- Board authorization and delegation to the Treasurer.
- Compliance with external laws and regulations (e.g., SEC rules 10b-18 and 10b5-1).
- That there was only one active agent on any given day that shares were repurchased.
- That a signed agreement was in place for each agent.
- For the plan repurchase program, there was only one plan in place at a time.

TREASURY OPERATIONS AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 13. Bank Guarantee

Objective:

Review the bank guarantee portfolio to validate that all guarantees are properly supported and monitored. It should include the issuance procedures, fees analysis, and business need validation.

Risk/Exposure:

Ineffective monitoring of bank guarantees could result in unnecessary financial exposure and costs.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 13. Bank Guarantee

Primary Test: Bank Guarantee Management (BG001.P)

Objective:

Review the bank guarantee procedures, portfolio monitoring, and fees to validate that all guarantees are properly supported, the portfolio is monitored by the management team, and their fees' analysis and business continued need validation were timely performed.

Approach:

Request the list of all bank guarantees managed during the audit period, including new issuances, or finished guarantees. Additionally, request the last continued business need portfolio review and the related approvals. From that universe, review a sample of bank guarantees to validate:

- The issued bank guarantees were properly supported, reviewed, and approved.
- The guarantee details match the approved contract and business requirements.
- The open-ended guarantees are being properly monitored, including their continued business need validation.
- A process exists for a periodic portfolio review, including the appropriate outstanding and expired bank guarantees.
- If the fees were properly charged as the fee schedule, and when they should stop (e.g., maturity date, physical return of the guarantee etc.).
- The outstanding guarantees were included in the monthly tracking report validated by management.

TREASURY OPERATIONS AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 14. Separation of Duties

Objective:

To ensure proper separation of duties (SOD) exists so that no individual can perform all control activities within a process in order to misuse or divert company funds without detection.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 14. Separation of Duties

Primary Test: Separation of Duties (SD001.P)

Objective:

To ensure proper separation of duties (SOD) exists so that no individual can perform all control activities within a process in order to misuse or divert company funds without detection. Essentially, one individual should not be permitted to perform activities which combine custody, transacting and recording of IBM activities or assets without a secondary review.

Approach:

- Request the organization's SOD assessment and review the responsibility assigned in the areas of treasury operations and accounting for potential SOD exposures as, but not limited to, the examples below:
 - Individuals responsible for transfer of funds and also responsible for their approvals or bank reconciliations.
 - Individuals responsible for trading/valuation of investments and also responsible for confirmations, cash transfers, recording of transactions; individuals responsible to transfer/receive funds for investment purpose and also responsible for reconciliations.
 - Individuals responsible for foreign exchange transactions cannot validate contracts, authorize movement of funds, approve or recording transactions, perform reconciliations of account in foreign currency.
- Does the assessment contain all duties (trading, recording, accounting, and reconciling) which are performed in the function? Were all conflicts identified in the assessment and the mitigation procedures documented? There must be a clear delineation of the responsibilities between the front, middle, and back offices. In addition, there should be physical separation between these functions.
- Specifically review bank signatories and the individuals who have responsibility to start or approve cash transaction, handle checks, perform bank reconciliations, write journal entries, and are involved in any payments or receipts operations.
- Validate the periodic review and approvals as required by FIN182 Appendix C.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 15. Business Continuity Planning

Objective:

To determine if process owners have developed, reviewed, and tested business continuity plans (BCP) so that in the event of natural or man-made disasters or incidents, pre-defined plans can be executed to maintain critical business operations.

Risk/Exposure:

Any material disruption of business operations due to natural or man-made disaster or incident could lead to significant financial loss not only IBM but potentially also for IBM's customers. In the treasury operations process it could also lead to an inability to ensure that treasury transactions or critical payments (e.g., payroll and taxes) are performed. Many business processes have been centralized, and often operations are conducted in locations which may be susceptible to business disruption. As a result, it is important that business and process owners have developed plans to maintain continuity of business operations in those centralized operations.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 15. Business Continuity Planning

Secondary Test: Business Continuity Planning (BC001.S)

Note: Business continuity risks become increasingly important in the environment where there are centers of excellence and multiple countries are supported by one center. This should therefore be an important test and considered primary in that environment. IT testing for disaster recovery is handled by the IT audit teams.

You should refer to BPBC127 Guideline for Business Process - Area Business Continuity Planning in the reference section of this audit program, in support of your review.

Objective:

To determine if process owners have developed, reviewed, and tested business continuity plans (BCP) so that in the event of natural or man-made disasters or incidents, pre-defined plans can be executed to maintain critical business operations.

Approach:

A guideline has been put in place for process owners providing guidance on business continuity planning (BCP). It should be noted by the audit team that this is a guideline rather than a specific instruction. However, it will act as a guide for you to understand whether key risks have adequately been considered for BCP and enable you to make an informed judgement regarding the risk mitigation in place. Different processes will have their own unique risk profiles regarding the probability and impact of an interruption to business operations. For example, the probability and impact of a business interruption for a process which has been centralized for multiple countries in a location with less-developed infrastructure is likely greater than that of a process supporting only a single country in location with robust infrastructure. As such, auditor judgment will be required to determine how important BCP is for a particular process, as well as to determine the reasonableness of any plans that have been designed by the process owners.

In the initial data request information ask for the process owner to confirm what plans have been implemented to ensure business continuity in the event of a natural or man-made disaster. Review that non-IT process elements have been adequately considered in the BCP.

The following list of questions has been designed based on the BPBC127 Guideline for Business Process - Area Business Continuity Planning.

1. Has management assessed the risk to business continuity (BC) and identified the key business processes and activities that require formal business continuity planning based on impact to the business.
 - Identification and prioritization of business processes' activities in addition to prioritization done through by the Enterprise Process Owner (level 3 plans).
 - Understanding and documentation of the potential impact (financial, operational and reputation) to IBM from business disruptions from uncontrolled, non-specific events.
 - Identification of the legal and regulatory requirements for the institution's business functions and processes.
 - Review and approval from senior management.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 15. Business Continuity Planning

Secondary Test: Business Continuity Planning (BC001.S)

(Continued)

2. For critical business areas and processes, has the maximum acceptable outage (MAO) been established and considered in the BC plan?
 - Estimation of maximum allowable outage (MAO), based on the “acceptable level” of losses, associated with the institution’s business functions and processes.
 - Estimation of recovery time objectives (RTOs), recovery point objectives (RPOs), and Maximum Tolerable Downtime (MTD) of applications and supporting technology required for business operations.
 - If applicable, recommended critical path recovery order.
 - Reconciled with agreed IT recovery priorities (Business Criticality Values, RTO/MTD).
3. For business areas and processes where there is a business continuity risk, has the BC Plan been created, reviewed, and approved by management?
 - The organization shall establish documented procedures for responding to a disruptive incident and how it will continue or recover its activities within a predetermined timeframe.
 - Top management shall review the organization’s BCP, at planned intervals, to ensure its continuing suitability, adequacy, and effectiveness.
4. Does the Plan document the specific continuity and recovery actions for all four impact scenarios (workplace, workforce, IT, and regional) that will need to take place to mitigate risk to the business if the impact lasts longer than the documented business’ maximum acceptable outage?
 - Flexible to respond to unanticipated threat scenarios and changing internal conditions.
 - Focused on the impact of various threats that could potentially disrupt operations rather than on specific events.
 - Effective in minimizing service disruptions and financial loss through the implementation of mitigation strategies.
5. Does the BC plan clearly identify dependencies such as site resources, IT, suppliers, process interdependencies, partners, non-IT services and been reviewed and synchronized with the appropriate crisis management team?
 - An effective BCP coordinates across its many components, identifies potential process or system dependencies, and mitigates the risks from interdependencies.
6. Is there an established, management-endorsed business continuity team to oversee the maintenance, invocation, and execution of the BC plan?
 - Defined roles and responsibilities for people and teams having authority throughout an incident.
 - Additional teams as necessary to facilitate recovery execution per the chosen strategy.
7. Are personnel aware and educated on the plan, strategy, and roles?
 - Business continuity training should be an integral part of the Business Continuity Management.
 - Comprehensive training program should be developed for all employees, conducted at least annually, and kept up to date to ensure that everyone understands their current role in the overall recovery process.
 - An audit trail should be maintained to document management's training efforts.

TREASURY OPERATIONS

AUDIT PROGRAM #12-00-00

CONTROL ASSESSMENT QUESTIONS

Key Control Question: 15. Business Continuity Planning

Secondary Test: Business Continuity Planning (BC001.S)

(Continued)

8. Is there a regularly executed process for validating the Plan through tests including key elements such as people, technology (data, systems, applications, and telecommunications), and locations that house staff and technology to ensure fit for purpose and alignment with current risks?
- Tests should be based on appropriate scenarios that are well planned with clearly defined aims and objectives that promote continual improvement.
 - Produce formalized post-exercise reports that contain outcomes, recommendations, and actions to implement improvements and are reviewed with management.
 - Tests are conducted at planned intervals and when there are significant changes within the organization or to the environment in which it operates.
9. FOR COMMERCIAL CONTRACTS: Where IBM has customer contracts including business continuity planning, have BC plans been implemented in accordance with contractual obligations?

FOR CUSTOMERS WITH REGULATORY REQUIREMENTS: Does the plan accommodate and account for the any unique rigor or accommodations for the regulatory body.

- Adherence with contractual obligations contained in the IBM Responsibilities section of the Schedule or similar agreement. Including communication, recovery parameters, specialized technical designs.
- Special considerations to meet unique regulatory requirements (HIPAA – US Healthcare has unique wording on backup, protection, and availability of data).
- Note: Usually this item is not applicable for treasury operations but should be part and evaluated as the other BCP items.